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Teaching Tools and Strategies

Schema Study: A Large Language Model (LLM) Application for Asynchronous Student Learning and Inquiry

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Abstract

Undergraduate biology educators face a critical challenge: providing immediate, personalized formative feedback to increasingly large, diverse classes. Large Language Models (LLMs) offer potential solutions, but open-ended chat interfaces pose challenges including curricular misalignment and equity gaps. We developed Schema Study, a free, no-code, open-source web application where instructors upload course terms and context via a single spreadsheet to create an AI-powered chatbot. Our LLM tutor uses evidence-based teaching practices and Socratic questioning to deepen understanding, correct misconceptions, and encourage students to find connections among course concepts. During Winter 2025, we integrated Schema Study into an introductory biology course, embedding it within structured assignments and updating content weekly. Pre- and post-surveys (N=225) indicated strong student satisfaction; 72% would reuse Schema Study in future biology courses. Each additional day per week students used Schema Study more than doubled the likelihood they would recommend it. Schema Study enhanced students' AI self-efficacy and their belief that AI is relevant to their education and careers. Through iterative, classroom-based refinement, we updated the application based on student feedback, highlighting best practices for integrating LLM chatbots: clear structured messaging, AI literacy training, curricular alignment, and scaffolded active learning opportunities. The tool provides formative practice through question-led dialogue; independent performance is evaluated in secure assessments outside the app. Schema Study offers a scalable, accessible strategy for biology educators to leverage generative AI's benefits while mitigating its risks.

Primary Image: A student studying on their computer at home. *Image generated by OpenAI GPT-4o. Accessed May 04, 2025.*

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INTRODUCTION

Large Language Model (LLM) systems, such as OpenAI's ChatGPT and Anthropic's Claude, have attracted widespread interest in higher education due to their ability to generate fluent, context-specific explanations on demand (1,2). Researchers have shown that LLMs can provide rapid formative feedback and support metacognition at scale (3). Effective feedback addresses three key questions: 'Where am I going?' (feed up), 'How am I going?' (feed back), and 'Where to next?' (feed forward) (4). Such formative feedback differs fundamentally from simply providing answers, as it requires students to revise, justify, and connect ideas through active response. However, open-ended chat interfaces may also introduce misinformation or distracting tangents unless carefully guided (5, 6). This highlights the need for intentional educational design when integrating LLMs into learning environments, with an emphasis on structured feedback, goal setting, and personalization (7). Biology educators, therefore, require integration strategies and practical teaching tools that enhance the benefits of LLMs while mitigating their costs and risks.

This need is especially pressing in introductory biology, where students must demonstrate mastery of terms and concepts and understand how they interconnect, often across levels of biological organization. Following Willingham (2019), we recognize that critical thinking in biology is discipline-embedded, requiring domain-specific knowledge for mechanistic reasoning, evaluating claims, and transfer to new contexts (8). Students who study actively, such as by self-quizzing, explaining, and connecting ideas, earn higher grades and maintain stronger academic self-efficacy than peers who rely on rote memorization (9-12). The benefits of explicit study skill instruction have been demonstrated among undergraduates who identify as English language learners (13), first-generation students (14-16), students from underrepresented minority groups (URMs) (17, 18), and neurodiverse students (19). Unfortunately, increasing class sizes and packed schedules limit the amount of support instructors can provide to each student. After the COVID-19 pandemic, asynchronous study aids became indispensable, yet most existing tools, including flashcards, simulations, and low-stakes quizzes, lack the technological capability to provide immediate, adaptive, and personalized formative feedback (20-23). While term retrieval

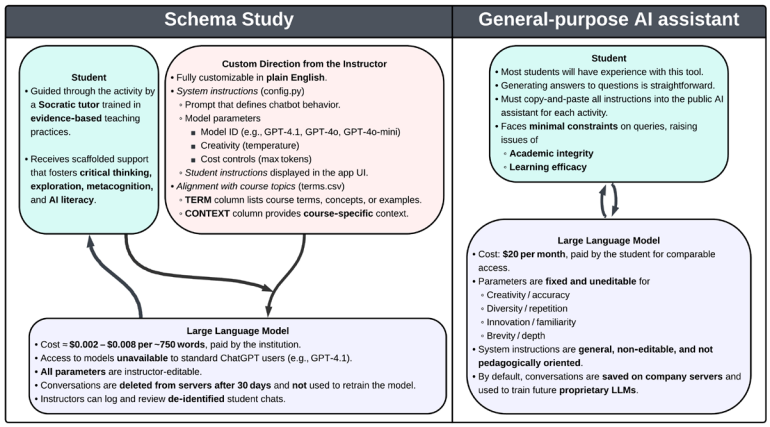

Figure 1. Schema Study (left) vs. a general-purpose AI assistant (right). Schema Study enables instructors to configure model settings, match prompts to course material, and keep conversations private. The institution covers all associated costs. Students interact using scaffolded Socratic dialogue. In contrast, general purpose AI assistants store conversations for proprietary LLM training by default, require students to pay monthly for access to advanced models, and prioritize user satisfaction over educational effectiveness.

through flashcards is useful, explanation and application are required for durable learning; effective tools must scaffold that deeper step.

LLMs have the potential to provide such formative feedback; currently, some match or exceed human performance on many biology knowledge assessments (24-26), and their conversational format naturally supports Socratic questioning. We define the Socratic approach as question-led dialogue that (1) withholds direct solutions while providing brief, targeted feedback on student responses, (2) poses exactly one scenario-grounded follow-up question per turn, and (3) presses for mechanistic reasoning, justification, and connections between concepts, drawing from evidence on elicitative questioning in science education (27). Yet four barriers hinder adoption in the undergraduate biology classroom:

1. **Lack of curricular alignment:** generic chatbots are not tuned to an instructor's learning objectives (28).
2. **Misalignment with evidence-based teaching strategies:** commonly used chatbots exhibit tendencies toward AI sycophancy and excessive information sharing, rather than employing instructional strategies such as scaffolding and Socratic questioning (7, 29).
3. **Equity and access:** many commercial AI study tools sit behind paywalls, exacerbating resource gaps (30).
4. **Technical burden:** biology educators without computer programming experience cannot easily customize or host AI apps without paying for proprietary services (31).

Schema Study: A No-Code, Instructor-Curated LLM Tutor

To address these challenges, we developed Schema Study, an open-source web application that turns a spreadsheet of course terms into an AI-powered study coach. We situate Schema Study alongside classroom-oriented AI tools such as Khanmigo (32), LearnLM/Gemini in Classroom (33), and ChatGPT Study Mode (34), while noting our focus on instructor control, course specificity, open-source access, and low cost. Instructors upload a two-column "TERM–CONTEXT" file, optionally edit plain-language prompt settings, and launch a web application that students can access from any browser. During a session, a student selects a term (e.g., natural selection), and the chatbot, named Pliny, initiates dialogue by asking the student to explain the term in their own words. Hidden system instructions direct the LLM to (a) give positive reinforcement, (b) correct misconceptions, and (c) pose progressively deeper questions while incorporating instructor-provided context. By ending responses with an open-ended question or a real-world scenario, Pliny helps students connect the focal term to other course-relevant biological concepts, terms, and examples (e.g., mutation, sexual selection, the electron transport chain, Jennifer Doudna). Figure 1 shows a schematic of how a typical student-chatbot conversation works within Schema Study compared to a general-purpose AI assistant such as ChatGPT or Claude.ai.

The design keeps instructors in control: they choose the LLM model (default = gpt-5.2), as well as the parameters and system instructions that guide chat interactions; they can update the terms-and-context spreadsheet weekly to match lecture topics; and they can modify the full system prompt without needing to write code. Because Schema Study is a free, no-code, open educational resource (OER), biology instructors can adopt it with minimal financial cost and computing infrastructure.

In service of our primary goal of creating a no-code AI tool that gives instructors maximum control, we use iterative classroom-based refinement aligned with design-based research principles to improve Schema Study based on student feedback and to draft best practices for the use of custom LLM tools in the classroom (35). In the sections that follow, we will describe Schema Study's architecture and how we implement it in the classroom, present evidence of student engagement, and offer best practices for integrating no-code LLM tools into undergraduate curricula.

SCHEMA STUDY APPLICATION STRUCTURE

Flashcards are among the most common study tools used by students, typically pairing a term with its relevant context. However, traditional flashcards primarily support rote memorization rather than deeper conceptual understanding (36). While term retrieval through flashcards is useful

for building foundational knowledge, explanation and application are required for durable learning. To address this limitation, we designed Schema Study to transform a simple list of course terms into an interactive, AI-powered learning experience that scaffolds students beyond definition toward mechanism, transfer, and claim evaluation. Students engage in meaningful, adaptive conversations with an AI chatbot named Pliny that combines the conversational capabilities of a large language model (LLM) with instructor-curated course content. Rather than passively reviewing static information, students actively explain concepts, receive immediate formative feedback as defined earlier—addressing 'Where am I going?', 'How am I going?', and 'Where to next?'—and explore connections among biological ideas, all at no cost to students and accessible anytime through a web browser (4).

Schema Study is structured as a no-code web application, intentionally designed to be user-friendly for biology instructors without programming experience. While other LLM tools exist for undergraduate biology learning, most are either proprietary and subscription-based or require computer programming experience to set up (Table 1). Within Schema Study, instructors configure the application using plain language rather than complex coding. Instructors tailor the tutor by editing two files: a terms and context spreadsheet that contains course content (Table 2, Supporting File S1) and a plain-English configuration file that guides the behavior of the LLM and the content of the web application, all without requiring any programming knowledge or experience (Table 3). Below, we describe the application's primary components.

Terms and Context Spreadsheet

At the core of Schema Study is the terms.csv spreadsheet, organized into two columns labeled "TERM" and "CONTEXT" (Supporting File S1). Instructors populate the "TERM" column with key concepts, skills, examples, or historical figures aligned with course learning objectives. Students select terms from a dropdown menu in the web interface to initiate interactive study sessions (Figure 2).

The "CONTEXT" column allows instructors to provide additional course-specific details that Pliny emphasizes during interactions. Context entries can range from brief notes to comprehensive explanations, related terms, practice questions, alignment with course materials, or links to external resources. This information is then provided to an LLM to create a chat-based conversation that is preferentially aligned with an instructor's course. This targeted customization enriches Pliny's responses, helping students integrate new information into existing knowledge schemas. For example, when adding context for "natural selection," an instructor might include course-relevant subtopics, scientist highlights, or URLs to validated online resources. Additional examples of effective context entries are provided in Table 2.

Application Setup and Configuration

Initial setup and testing of Schema Study can be done individually or through structured professional development workshops. We designed Schema Study to give instructors full control over the chatbot's behavior without requiring coding skills.

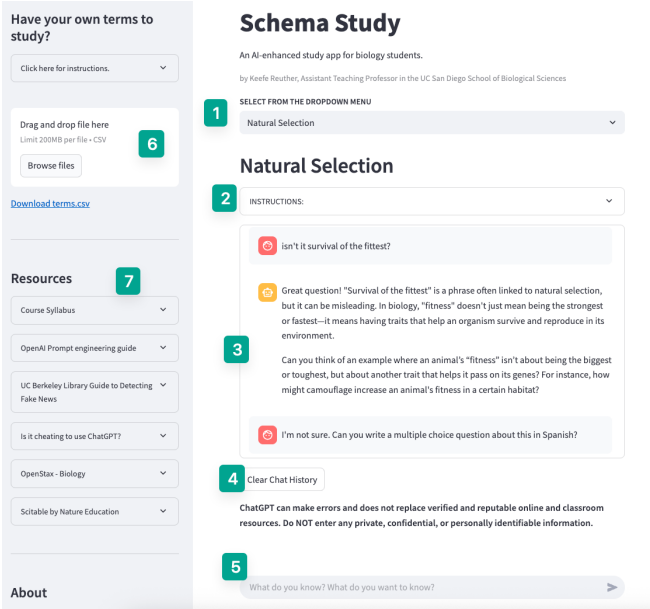

Figure 2. Schema Study user interface. 1. Students begin by selecting a course term from the drop-down menu; this choice anchors the entire dialogue. 2. The term appears as a heading, and an expandable panel displays in-app instructions. 3. The chat window shows the unfolding Socratic conversation between the student (pink) and the AI tutor Pliny (gold). 4. A Clear Chat History button lets users reset the session and start over with a new term. 5. The input bar invites students to type questions, explanations, or code snippets. 6. Instructors—or students—can drag-and-drop a customized terms.csv file to swap in their own vocabulary list. 7. A collapsible sidebar links to the course syllabus, prompt-engineering tips, information-literacy guides, and other instructor-curated references. This represents the interface in version 1 – the version used by students in Winter quarter 2025.

For instructors interested in testing the application prior to creating their own copy, two preview versions are available on Hugging Face Spaces. The original version used by students in Winter 2025 can be accessed here, and the current updated version incorporating student feedback is available here. Note that preview versions require users to paste in their own API key; see Supporting File S2 for instructions on obtaining one.

The production version of Schema Study, designed to be copied and customized for use with students, is available here. Supporting File S2 provides a complete walkthrough of copying, customizing, and hosting a custom version. Please email the corresponding author with any questions. The full list of configuration options is provided in Table 3 and Supporting File S2.

LLM Choice and Payment

There is a large array of LLMs that can be integrated into a custom chatbot. These can range from open-source models (e.g., Llama4, Mistral) to proprietary models (e.g., OpenAI's GPT, Anthropic's Claude, Google's Gemini). Most of these can either be run from personal/institutional servers or servers controlled by private corporations. Running any LLM locally that can sustain many high-quality, simultaneous, real-time chat conversations often requires prohibitively expensive computing power. Therefore, the best option is to choose a privately controlled LLM that is paid for with an application programming interface (API) key. An API key is a unique identifier, essentially a secure digital password, that allows

software applications to communicate directly with external services, such as LLMs. This does require having clear dialogue with your students about the privacy of their data and avoiding entering in any personally identifiable or protected data.

We chose OpenAI over other providers for two reasons. First, since the computing code underlying the app is specific to the provider, we had to choose a single company. We felt that offering multiple LLM choices would make Schema Study unnecessarily complex. Second, when Schema Study was initially developed, OpenAI and Anthropic were the most affordable and possessed the most robust documentation, functionality, and user support. We settled on OpenAI since that is the framework we were already most familiar with. However, since the code for Schema Study is open source, we encourage others to develop versions that can be used with non-OpenAI LLMs.

When you use Schema Study or similar custom chatbots, the application sends requests to an external AI provider (e.g., OpenAI) to generate responses. The API key authenticates these requests, ensuring secure and authorized access to the AI service. Using API keys is both necessary and cost-effective for several reasons:

- API keys ensure that only authorized users or applications can access the AI service, protecting your account from unauthorized use.
- API keys allow providers to track usage accurately, enabling transparent billing based on actual use rather than flat subscription fees.
- Instead of paying for expensive subscriptions or licenses, institutions pay only for the AI resources they actually use. For example, Schema Study's implementation with approximately 300 students over 11 weeks cost our institution just $177.16—far less than individual student subscriptions would have cost.
- The financial burden is transferred from the student to the institution, which reduces equity gaps associated with the use of LLMs (37).

In short, API keys provide a secure, flexible, and economical way for educators to integrate powerful generative AI tools into their teaching without incurring unnecessary expenses or complexity. Importantly, the app does not retain chat transcripts; evidence of learning is produced via LMS-submitted screenshots and reflections, ensuring privacy by design.

Application Configuration

Instructors can customize both the in-application instructions visible to users and the system instructions sent to the LLM, all without needing any coding expertise. All essential settings are configured using plain language within a single file: config.py. Supporting File S2 provides a detailed explanation of each parameter, along with instructions for editing them. Below, we discuss two key settings that most significantly impact the user experience: model choice and system instructions.

Model Choice

Schema Study allows instructors to choose from multiple AI models, each with distinct performance characteristics and associated costs. Selecting a model involves balancing factors such as response quality, complexity, speed, and cost efficiency. The model choice is configured as a single parameter in the configuration file, facilitating easy updates as newer models become available. As of the writing of this paper, Schema Study defaults to GPT-5.2, which offers a strong balance of accuracy, responsiveness, and affordability. Instructors may also select GPT-5.1 or GPT-4.1 depending on their preferences for cost, speed, and adherence to system instructions. The authors will update models and parameters in the future when either tools become deprecated or significantly improve Schema Study. Please check the corresponding author's Hugging Face page for the most current version of Schema Study.

System Instructions

The chatbot's behavior is guided by a carefully crafted set of instructions embedded in each interaction. These instructions define Pliny's role, conversational style, and pedagogical approach, ensuring alignment with course objectives. During Schema Study's development, we used iterative classroom-based refinement aligned with design-based research principles to balance structured guidance with flexibility for diverse student inquiries. The current version's system prompt enforces one focused question plus one short, realistic application example per turn to reduce confusion from multiple simultaneous follow-ups. Although instructors may modify the system prompt, we recommend making only small, incremental adjustments to ensure the chatbot maintains its intended functionality. After each change, we advise thorough testing to assess how modifications influence the model's responses, as even minor tweaks can significantly impact the chatbot's output and effectiveness in addressing student questions (38).

Core Principles of Schema Study:

1. **Promote active learning and critical thinking:** Encourage students to articulate their understanding before providing guidance, progressively deepen their conceptual knowledge, and help them build interconnected schemas of biology concepts.
2. **Provide supportive and constructive feedback:** Offer positive reinforcement, gently correct misconceptions, and foster a growth mindset through supportive interactions.
3. **Communicate clearly and concisely:** Use simple, jargon-free language, keep responses brief to manage cognitive load, and introduce information step by step to enhance comprehension.
4. **Engage through Socratic dialogue aligned with course context:** End responses with open-ended questions and relevant real-world scenarios, prompting students to explore connections among biological concepts and course-specific examples, using our defined Socratic approach of question-led dialogue that withholds solutions while scaffolding deeper understanding.
5. **Maintain professionalism, inclusivity, and focus on relevant topics:** Communicate professionally and inclusively, use culturally relevant examples, and politely redirect irrelevant inquiries, focusing exclusively on biology topics pertinent to academic success.

IMPLEMENTATION AND RECOMMENDATIONS

For Students in the Classroom

Classroom Context and Integration

From January to March 2025, in a large undergraduate biostatistics course, we introduced Schema Study to students. Designed primarily for lower-division biology majors, this course teaches experimental design, statistical analysis, and basic computer programming using the R language. We incorporated Schema Study as a tool throughout the course to strengthen students' understanding of course concepts, enhance their critical thinking skills, and train them to use LLMs effectively. While not initially designed with Willingham's discipline-embedded critical thinking framework in mind, we now recognize that Schema Study's structure aligns well with this framework (8). The `terms.csv` file (S1) provides the domain-specific knowledge base that Willingham identifies as necessary for meaningful critical thinking in biology, enabling students to move beyond surface-level memorization to mechanistic reasoning and transfer. All activities were conducted as regular course instruction with student-submitted evidence only under IRB protocol #807175.

We introduced the tool through a live demonstration during lecture, highlighting its features and benefits. We also provided an overview of generative AI, setting clear expectations for academic integrity and appropriate use. Additionally, we designed two low-stakes online discussion board assignments hosted on the university's learning management system (LMS) that required students to use Schema Study at varying levels of complexity. To ensure ongoing relevance, we updated the `terms.csv` spreadsheet (S1) weekly to align with current course topics and reinforce newly acquired knowledge. Our goal was for students to choose to use the application throughout the course to prepare for other assessments such as coding assignments, quizzes, and exams.

Designed Classroom Activities

Two key activities were developed to encourage engagement with Schema Study. The full text can be found in Supporting File S3, which includes example student responses and mastery rubrics for each activity. Across these activities, evidence of learning is captured through four types of mastery artifacts: (1) student-written explanations of focal terms, (2) cross-term applied scenarios (short, realistic situations), (3) claim-verification tasks where students evaluate AI output against sources for inaccuracies or hallucinations, and (4) brief reflections on how feedback changed their thinking. Each activity emphasizes different artifacts: Activity 1 focuses on claim verification and information literacy, while Activity 2 emphasizes connecting concepts through applied scenarios. Both activities require students to engage with targeted prompts, revise their understanding based on feedback, and submit reflections to the LMS.

Activity 1: Exploring Challenging Topics and Verifying Information

Students were tasked with selecting a term from Schema Study that they found confusing or intriguing. They initiated conversations with the chatbot to explore the term in depth and encouraged the AI to provide detailed explanations. Students then identified an interesting or unexpected piece of information from the dialogue and took a screenshot of that snippet.

In their discussion board posts on the LMS, students included the screenshot and wrote one to two sentences explaining their choice. They were instructed to focus on information that required citation or was not common knowledge. Classmates were then required to respond to a peer's post by conducting independent research to verify the chatbot's information using reliable, scholarly sources. They provided APA-formatted citations and brief explanations confirming or refuting the chatbot's claims. This activity promoted the critical evaluation of AI-generated content and reinforced information literacy skills.

Activity 2: Connecting Concepts Through Hypothetical Scenarios

In the second activity, students selected a term they felt confident explaining. They then prompted the chatbot to help them connect it to another randomly chosen term from the term list. The chatbot assisted the student in creating a hypothetical real-life scenario linking the two terms and provided constructive feedback. Below is an example of the prompt we instructed students to give Schema Study, along with the chatbot's response, illustrating how Schema Study scaffolds discipline-specific reasoning through turn-by-turn Socratic dialogue.

> *Initial Prompt: "I want to test my ability to connect this term to others in the term list. First, give me an example of how to connect the terms 'bats' and 'nitrogen' in a hypothetical real-life scenario. Second, prompt me to similarly create a logical applied scenario between the displayed term and one other you MUST CHOOSE RANDOMLY from the course term list. Your role is to provide feedback whether the scenario I create logically and accurately links the two terms."*
>
> *Pliny: "Great initiative! Here's an example of connecting 'bats' and 'nitrogen': Imagine a tropical forest where bats feed on insects. The bats' droppings (guano) accumulate on the forest floor, enriching the soil with nitrogen. This added nitrogen helps plants grow more vigorously, which in turn supports a richer ecosystem. So, bats indirectly influence the nitrogen cycle in their habitat. Now it's your turn! Can you create a real-world scenario that connects 'Natural Selection' (our displayed term) and 'Galapagos Tortoises' (randomly chosen from your list)? Try to describe how these two terms might interact or be relevant together in a biological context. I'll give you feedback on your scenario!"*

This example demonstrates how Schema Study can provide the prerequisite domain knowledge (understanding of nitrogen cycling, ecosystem interactions) that Willingham (2019) identifies as essential for critical thinking. The step-by-step progression from concept identification to application in realistic contexts exemplifies discipline-embedded reasoning in practice. Students engaged thoughtfully with the chatbot and crafted scenarios that demonstrated a deep understanding of both concepts. They were instructed to avoid superficial connections and instead focus on meaningful, creative applications. In their discussion board submissions, students included the terms used, their hypothetical scenario, a screenshot of the chatbot's

formative feedback, and a short reflection on the exercise, noting whether they agreed with the feedback. By requiring students to articulate connections between concepts and engage in peer review, the exercises supported the construction of integrated knowledge structures essential for mastering complex biological topics. We hope that by informally gamifying the experience, we can increase engagement, particularly among students who lack strong study skills (39). Independent performance is evaluated in secure assessments outside the app (e.g., proctored exams, timed quizzes), maintaining a clear boundary between formative practice in Schema Study and summative evaluation.

Student Survey Methods and Results

We invited all students enrolled to complete pre- and post-course surveys delivered through Qualtrics. Of the 288 enrolled students, 225 submitted both surveys (response rate = 78.1%; Supporting File S4).

The proportion of students who would choose to use Schema Study again for another biology course (71.56%) was higher than the proportion who would choose to use it in a non-biology STEM course (63.11%) or a non-STEM course (44.44%). Since LLMs are better known for their strengths in coding and writing, this increased preference for usage in a biology classroom may reflect the positive effect of context and experience on students' perception of where LLMs are useful in their education. If Schema Study were used in a computer science or writing course, students might be more likely to recommend its use in those contexts.

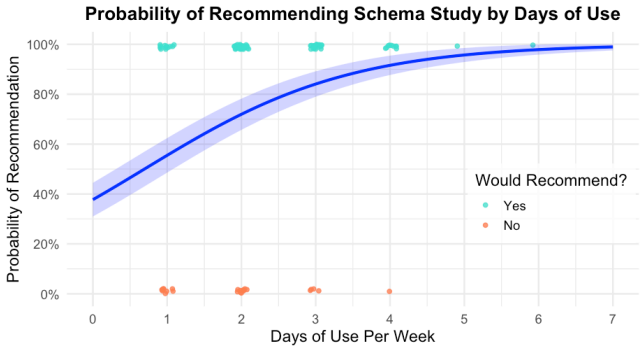

Figure 3. Logistic regression model showing the probability of recommending Schema Study based on frequency of use. Scatter plot with jittered points representing individual students (N = 199) plotted by self-reported days of use per week (0–7 days) and their binary recommendation response (Yes/No). Logistic regression analysis revealed that each additional day of use significantly increased the likelihood of recommending Schema Study ($\beta = 0.722$, SE = 0.213, $z = 3.391$, $p < 0.001$). The odds of recommending Schema Study doubled with each additional day of use per week (OR = 2.06, 95% CI: 1.39–3.20). The blue line represents the fitted logistic regression curve with 95% confidence interval (shaded area). Chi-square test of independence confirmed a significant association between frequency of use and willingness to recommend ($\chi^2 = 17.212$, df = 5, $p = 0.004$). This pattern indicates that moderate, regular engagement with Schema Study (frequency of use) predicts student satisfaction better than prolonged individual sessions (session length), suggesting the value of consistent, brief interactions with the Socratic dialogue format.

We examined whether self-reported usage (average days of usage per week and average minutes per session) predicted whether students would choose to use Schema Study in another biology course if given the opportunity. Students generally interacted with Schema Study one to two days per week, and a typical session lasted about thirty minutes (median = 24 min, IQR = 12–36). We used binomial logistic regression to test whether increased app usage significantly predicted students' likelihood of choosing Schema Study in the future for another biology course. Using R (v4.3) with the packages *dplyr, ggplot2, stats::glm, and broom,* the logistic model containing both predictors revealed that individual session length was not significant ($\beta = -0.003 \pm 0.011$ SE, $z = -0.26$, $p = 0.80$; OR ≈ 1.00), whereas days of usage per week remained significant ($\beta = 0.75 \pm 0.23$ SE, $z = 3.19$, $p = 0.001$). Each additional day per week students used Schema Study more than doubled the likelihood they would recommend its use (Figure 3; odds ratio = 2.06, $p = 0.0007$). Taken together, the data indicate that moderate, regular use—not prolonged or intensive study marathons—is what students perceive as most beneficialfinding suggests that consistent, brief engagement with the Socratic dialogue format is more valuable than extended single sessions.

A multiple linear regression predicting final course grade from days of usage per week, mean session length, and whether students would use Schema Study in a future biology course explained little variance (adjusted $R^2 = 0.015$; $F(3, 195) = 1.99$, $p = 0.12$). Days per week was again non-significant ($\beta = 0.012$, $p = 0.98$), whereas longer sessions were associated with a small decrease in grade ($\beta = -0.046$ per minute, $p = 0.033$).

Students' perceptions of AI, related self-efficacy, and its utility in their academic and career lives changed significantly over the 11-week academic quarter (Figure 4). Supporting File S4 provides a list of statements that students responded to using a 4-point Likert scale (1 = strongly disagree to 4 = strongly agree). We calculated the change in score for each student between the pre- and post-surveys. We generated 10,000 bootstrap resamples to obtain 95% confidence intervals and subjected the set of fourteen one-sample t-tests (H_0: $\Delta = 0$) to a Benjamini–Hochberg false discovery rate correction $\alpha = 0.05$). All calculations were performed in Python 3.11 with the libraries *pandas, numpy, scipy, and statsmodels*.

Students not only expressed stronger endorsement of the prospective benefits of generative AI—citing greater time-saving potential, richer feedback, novel insights, and clearer relevance to their future careers—but also heightened their recognition of possible drawbacks, such as over-reliance and impediments to collaborative skill development. The only item that decreased was the belief that generative AI helps "others more than myself," indicating that participants now view these tools as comparably valuable to their own learning.

Representative student comments included:

Perceived Strengths:

- *"I believe it was helpful as it focused specifically on the course content."*
- *"I liked Schema Study more than other AI programs like ChatGPT because of the personalized feedback and questions that sparked critical thinking. I felt more like I was actually learning something than just getting an answer spit back out at me."*
- *"When completing my final project I found Schema Study beneficial in correcting some simple errors in my code."*
- *"Helping me understand the meaning of specific code."*

Perceived Weaknesses:

- *"TBH I mostly just used general ChatGPT instead of Schema Study because I kinda preferred it but I think it gave more comprehensive answers."*
- *"The forced Socratic nature of the chatbot was frustrating, especially if I didn't know what questions to ask."*
- *"Schema study was valuable overall, but sometimes I would try to ask it something and I know the point is to get you to think, but I answered 15 of its questions once and it still didn't help me with what I asked, which frustrated me."*
- *"I never used Schema Study except for the discussion board, so I don't have an impression for this."*

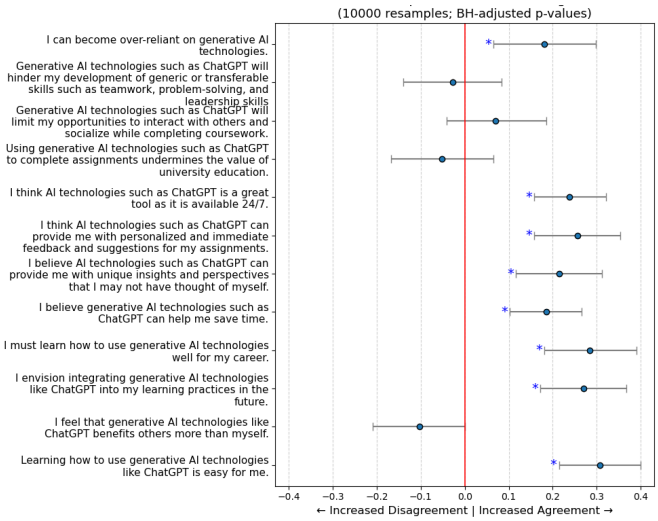

Figure 4. Changes in student perceptions of generative AI after using Schema Study. This forest plot displays mean change scores across survey items, with 95% bootstrap confidence intervals calculated from 10,000 resamples. Points represent mean changes in agreement with statements about generative AI tools following a quarter-long Schema Study experience (N = 225). Error bars show 95% confidence intervals derived from bootstrapping. The red vertical line marks no change (zero). Positive values indicate increased agreement; negative values indicate increased disagreement. Asterisks (*) indicate statistically significant changes from zero based on one-sample t-tests ($p < 0.05$), corrected for multiple comparisons using the Benjamini–Hochberg method.

Supporting File S4 lists all survey questions as well as additional figures and results.

Schema Study Improvements

Ongoing refinement is fundamental to design-based research, and Schema Study exemplifies this approach. Feedback and assessment drive the evolution of its design and the development of generalizable best practices (35).

All data, figures, and tables presented here reference Schema Study Version 1 (Evaluation Version 1; Student Version 1). After analyzing survey results, a consistent theme emerged: students wanted a simpler, more intuitive interface and increased guidance for getting started with a Socratic dialogue. These reports of confusion from multiple simultaneous follow-up questions motivated our change to the one-question pacing structure. In response, we have released multiple updated versions (Evaluation Version 2; Student Version 2), which introduced the following enhancements:

1. **Simplified Interface:** Reduced on-screen text for a more focused, user-friendly experience.
2. **Prompt Template Buttons:** Five customizable buttons now serve as conversation starters. Instructors can easily add, remove, or edit these (Supporting File S2):
   i. **Misconception Check:** Multiple choice questions addressing common misconceptions of the chosen term.
   ii. **Two Truths and a Lie:** A game requiring students to identify the lie about the chosen term and justify their choice.
   iii. **Connect Terms:** Aligned with Student Activity #2, this prompt guides students to form an applied scenario connecting two course terms.
   iv. **Schema Map:** Prompts Schema Study to find all terms in terms.csv that directly connect to the chosen term.
   v. **Create a Study Plan:** Helps students develop a study plan for an upcoming assessment.
3. **Removed Custom Uploads:** The ability for students to upload a custom terms.csv was retired due to low usage.
4. **Enhanced Pedagogy:** The revised terms.csv (S1) content and system instructions better support a consistent, Socratic-style tutoring experience (40).
5. **Web Search Enabled:** The instructor can decide whether the chatbot incorporates internet searches. This is particularly useful for finding accurate URLs of relevant resources or accessing knowledge created after the LLM model was released.

Both versions remain available as open educational resources. Supporting File S2 serves as the comprehensive user guide. We encourage educators to explore both and select the version that best supports their instructional goals. Our lean technology stack (Streamlit, Python, OpenAI API) with model choice as a single configuration parameter and prompt versioning on Hugging Face helps ensure long-term operability, though newer reasoning-oriented models may require minor prompt updates in the future.

Best Practices for Integrating LLMs in the Classroom

Drawing from both our experience and current literature (7, 41), we recommend the following practices for effective LLM integration:

1. **Set Clear Expectations:** Discuss LLM use with students and define policies in your syllabus.
2. **Differentiate Schema Study:** Emphasize that Schema Study is a guided, Socratic tool—not a generic chatbot like ChatGPT. Explore the pedagogical strengths and limitations of each. Instructors can use Schema Study to directly teach students about the role of traditional flashcards in learning (very useful, but not sufficient).
3. **Cultivate a Supportive Learning Ecology:** Use hands-on, low-stakes assessments that reveal where LLMs are helpful and where they fall short. Activities may include:
   i. Verifying the credibility of information.
   ii. Practicing self-reflection and study strategies.
   iii. Engaging in deep, Socratic dialogue.
   iv. Strengthening confidence through coding.
4. **Offer Consistent Support:** Reinforce learning through office hours and discussion sections.

5. **Continuously Tune and Test the Tool:** Even minor tweaks to system prompts can significantly affect behavior. Test thoroughly.
6. **Separately Evaluate Learning:** Maintain clear boundaries between formative practice (in-app) and summative evaluation (secure, out-of-app assessments). Use surveys to measure student experience and satisfaction with the tool, while using traditional assessments to evaluate actual learning outcomes. This dual approach recognizes that student satisfaction with AI tools does not necessarily correlate with learning gains, and that both metrics provide valuable but distinct insights for improving educational practice.

When thoughtfully implemented, Schema Study can increase student engagement and build foundational skills in critical thinking and independent learning. However, the tool's limitations must be acknowledged: it is less suited for high-stakes grading, original data analysis or code execution, wet-lab or psychomotor skills, team-process evaluation, creative writing or artistic expression, and mathematical proof construction.

Professional Development for Faculty

Schema Study debuted at the 2024 Society for the Advancement of Biology Education Research (SABER) national conference in a three-hour session titled "Developing No-Code, AI-Enhanced Online Study Tools for Biology Students: A Beginner's Workshop." Seventeen biology instructors from a range of institutions copied, customized, and tested their own versions of the app without any prerequisite programming skills. Workshop materials—the installation manual, slide deck, and step-by-step build guide—are available for independent use (Supporting Files S1, S2, S5, S6)).

Pre- and post-workshop survey data (Supporting File S6) highlighted changes in attitudes toward AI in education and opinions of the workshop and Schema Study. Of the 17 participants, 12 completed the pre-workshop survey, reflecting a diverse range of experience with LLMs.

Three primary themes emerged from the workshop that align with other AI-related workshops and seminars we have facilitated:

1. **Structured guidance is critical.** Many attendees requested a multi-session series or one-on-one follow-ups to solidify their first deployment.
2. **Time and technical bandwidth remain barriers.** Even with no-code tools, instructors cited limited time, institutional policies, and unfamiliarity with API billing as hurdles.
3. **Early adopters can catalyze wider uptake**. Participants comfortable with technology expressed willingness to mentor colleagues, echoing diffusion-of-innovation models in faculty development (42).

At our institution, we envision rolling out Schema Study and other AI-based pedagogical resources through three key phases: demonstration seminars to generate interest, hands-on workshops to help instructors create customized versions of the tool, and ongoing technical support and development to foster collaboration and research through discipline-based education research (DBER).

CONCLUSIONS

Schema Study is a scalable and accessible tool that promotes critical thinking through personalized, LLM-enhanced interactions. By providing a customizable and accessible platform, it adapts to the needs of students, instructors, subject material, and institutions—all without requiring extensive technical expertise or significant financial investment. Instructors can position flashcards as a starting point for retrieval, then use Schema Study to scaffold explanation, concept connections, and claim verification, thereby teaching students that while traditional flashcards are useful for term mastery, they are not sufficient for deeper understanding. Additionally, using Schema Study to interact with an LLM, rather than engaging directly with general AI assistants such as ChatGPT or Claude, ensures that identifiable data are not collected and chat messages are not used to train future models. As per OpenAI's privacy policy, all chat messages from Schema Study are deleted from OpenAI servers within 30 days and are not used to train its models (OpenAI).

Schema Study maintains clear boundaries between formative practice and summative evaluation: the app provides formative feedback through Socratic dialogue, while independent mastery is assessed through secure, out-of-app summative assessments. This separation ensures that the tool supports learning without compromising assessment integrity.

LLM technology is poised to advance rapidly and become an integral part of education, much like the internet over the past 30 years. To ensure that such technology equitably supports pedagogical goals, it is crucial to pair accessible professional and curricular development with rigorous, discipline-based education research (DBER). Empirical testing is needed to assess tools like Schema Study for their impact on learning outcomes and attitudinal characteristics such as self-efficacy, intrinsic motivation, and metacognitive skills, particularly within the framework of iterative classroom-based refinement aligned with design-based research principles (35, 43).

It can be challenging to attribute learning gains to a general course tutor, and so we decided against attempting it in this study. However, controlled experiments on learning in smaller, focused activities appear to show the positive impact of LLMs when used appropriately. Recent randomized controlled trials have provided compelling evidence for the efficacy of pedagogically-informed LLM chatbots in enhancing learning. Kestin and colleagues (44) compared a custom AI tutor with active learning classroom instruction in undergraduate physics, finding that students using the AI tutor learned significantly more in less time while also reporting higher engagement and motivation. Similarly, Kinder and colleagues (45) found that adaptive feedback generated by ChatGPT led to improved performance and increased engagement compared to static expert feedback among pre-service teachers. In second language learning, Zhang and Huang (46) demonstrated that LLM-based chatbots significantly aided students in acquiring both receptive and productive vocabulary knowledge, with particular benefits for long-term retention. These findings underscore the importance of empirical validation and suggest that instructor-controlled, course-specific AI tools like Schema Study merit rigorous assessment of their impact on learning outcomes across diverse educational contexts.

Future work should leverage Open Educational Resources (OER) and advancing technology to integrate tools like Schema Study with existing learning management systems. Such integration could provide valuable usage analytics within a framework that better safeguards students' FERPA rights and privacy (5). Additionally, adapting the platform to incorporate open-source AI models may lower costs and offer greater control over functionalities, addressing technical challenges associated with third-party APIs. Both research and technological innovation must also prioritize information literacy. While chatbots can perform remarkable tasks, they are not verifiable resources capable of linking statements to experimental evidence. This underscores the growing need to train students to evaluate the quality of information critically and recognize cognitive biases.

Expanding Schema Study beyond biology into other STEM and non-STEM disciplines could further enhance its impact. While much of the current research on LLMs in STEM education focuses on computer science, there is increasing interest in its applications within health sciences. However, studies on LLMs' role in other areas, such as biology, chemistry, physics, statistics, and scientific writing, remain limited. Collaborative efforts with educators across disciplines could help identify best practices and clarify how these practices differ based on specific learning objectives.

Addressing barriers to adoption is essential for widespread implementation. Comprehensive support resources—such as detailed user guides, tutorial videos, and technical assistance—can help educators overcome challenges related to time constraints and technical expertise. Establishing a community of practice where instructors exchange experiences, share best practices, and refine customizations can foster continuous improvement and innovation.

Adopting tools like Schema Study represents a meaningful step toward enhancing asynchronous and independent learning through AI-driven personalized interactions. By refining its features, broadening its applicability, and fostering collaboration among educators, we can harness the transformative potential of generative AI to improve learning outcomes and equity in education.

INSTITUTIONAL REVIEW BOARD (IRB) APPROVAL

This study was conducted under IRB protocol #807175, titled "Schema Study: Developing and Assessing a Web App with AI Formative Feedback to Enhance Student Study Skills and Learning." The protocol received initial exempt review approval on April 13, 2023, and the current expedited review approval on April 12, 2024.

SUPPORTING MATERIALS LIST

- S1. Schema Study – Terms.csv
- S2. Schema Study – User Manual.docx
- S3. Schema Study – Student Activities.docx
- S4. Schema Study – Student Surveys.docx
- S5. Schema Study – Teaching Slides.pptx
- S6. Schema Study – Workshop Surveys.docx

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Generative AI tools were used throughout the preparation of this manuscript, as well as during the design and testing phases of the Schema Study app. All content has been independently verified, and the authors accept full responsibility for its accuracy and integrity. The large language models (LLMs) used include:

Anthropic
- **Models:** Claude 3, 3.5, & 4.5 Series (Sonnet and Opus)
- **Role:** Assisted with copy editing and code generation

OpenAI
- **Models:** GPT-4 & GPT-5 Series (including Thinking mode); o-Series (o1, o3, o4-mini)
- **Role:** Assisted with copy editing, code generation, brainstorming, and drafting critiques

DeepSeek
- **Models:** DeepSeek R1, DeepSeek Coder V2
- **Role:** Locally hosted LLMs supporting research on protected data

Google
- **Models:** Gemini 2.5 & 3 Pro
- **Role:** Assisted with code generation and copy editing

Table 1. Comparison of Schema Study with other generative AI educational tools. Schema Study is a free, instructor-friendly, no-code generative AI web application compared with commercially available and open-source alternatives. The table compares 13 educational AI tools across five key dimensions: student pricing, coding requirements, open-source availability, instructor control level, and course-specific customization capabilities. Schema Study is the only tool that simultaneously offers all five desired features: free student access, no coding requirements, open-source availability, high instructor control, and full course customization. Note: Features and pricing as of August 2025; subject to change.

Name	Pricing for students	Coding required	Open source	Instructor control level	Course-specific customization	Reference
Schema Study	Free	No	Yes	High	Yes	https://huggingface.co/spaces/keefereuther/Schema_Study
Pearson AI Tutor	Pay	No	No	Medium	Limited	https://www.pearson.com/channels/biology/ai-tutor
McGraw Hill AI Reader	Pay	No	No	Low	Limited	https://www.mheducation.com/highered/digital-products/ai
CourseAssist AI	NA	No	No	High	Yes	https://courseassistai.com
Cognii	NA	No	No	High	Yes	https://www.cognii.com
Khanmigo (Khan Academy)	Pay	No	No	Low	No	https://www.khanmigo.ai/
LearnLM/Gemini in Classroom (Google)	Free	No	No	Low	Limited	https://edu.google.com/intl/ALL_us/ai/gemini-for-education/
ChatGPT Study Mode (OpenAI)	Free	No	No	Low	Limited	https://openai.com/index/chatgpt-study-mode/
Class Companion	Free	No	No	High	Yes	https://classcompanion.com
Edcafe AI	Free	No	No	High	Yes	https://www.edcafe.ai
SPOCK (Rice)	Free	Yes	Yes	High	Yes	https://github.com/luffycodes/Tutorbot-Spock
TeachGPT	Free	Yes	Yes	High	Yes	https://github.com/Priyamakeshwari/TeachGPT
Ruffle & Riley	Free	Yes	Yes	High	Yes	https://github.com/rschmucker/ruffle-and-riley

Table 2. Example entries from Schema Study's terms and context spreadsheet. Illustrates the type of structured content instructors provide to guide the Schema Study chatbot ("Pliny") interactions. This context may include comprehensive definitions, associated course lectures, common misconceptions, examples from class discussions, related terms, resources, career guidance, and important course policies. Supporting File S1 includes this text in a format readable by Schema Study.

Term	Context
Natural Selection	• **Course definition:** Natural selection is the process by which individuals with heritable traits that enhance fitness are more likely to pass on their alleles to the next generation. Over time, favored alleles and heritable phenotypes become more common. • **Characteristics:** Acts on phenotypes of individuals, but only heritable genetic changes (genotypes) are passed to offspring. Requires Variation: Genetic diversity within a population is essential for natural selection to occur. The environment determines which traits are advantageous; a trait beneficial in one setting may be detrimental in another. • **Misconceptions:** While natural selection favors traits that increase fitness, it does not work towards a specific goal or perfect organism. • **Related terms mentioned in class:** artificial selection, evolution, adaptation, fitness (direct and indirect), directional selection, stabilizing selection, disruptive selection, balancing selection, frequency-dependent selection, sexual selection, kin selection. • **Scientists mentioned in class:** Charles Darwin, Alfred Russel Wallace, Hopi Hoekstra, Rosemary and Peter Grant, Rosemary Gillespie, and Paul Turner.
Mutation	**Examples discussed in class:** Antibiotic resistance in bacteria, sickle cell anemia, Huntington's disease, CRISPR/Cas9 gene editing technology.
Sexual Selection	• **Associated lectures:** Week 3 Monday • **Associated assessments:** Quiz 3, Midterm 1, Final Exam, Discussion Board Post Week 4 • **Resources:** https://openstax.org/books/biology-2e/pages/19-3-adaptive-evolution
Gene Flow	[can be left blank]
Careers in Ecology and Evolutionary Biology	Point students to the following URLs: • Career Center: https://career.ucsd.edu/ • Outside Resources: https://openstax.org/books/college-success/pages/12-introduction https://www.colorado.edu/ebio/undergraduate/careers ; https://www.careerexplorer.com/degrees/evolutionary-biology-degree/
Course Regrade Policy	Any student can request a re-grade for any assessment. The request must be made within one week of the assessment being returned. The request should include a detailed explanation of why you believe the assessment was graded incorrectly. All requests should be in writing and emailed directly to the instructor.

Table 3. Schema Study customization controls via config.py. Key no-code configuration options available to instructors through the config.py file. All settings use plain-English instructions without requiring programming expertise. Complete parameter list and detailed editing instructions available in Supporting File S2.

Term	What it Controls	Example Options	Why Instructors Customize This
App Appearance	Application title and student warning messages	Change title to "Bio 101 Study Coach;" edit privacy disclaimers	Match course branding; address institutional policies
AI Model Settings	Which OpenAI model powers the chatbot, reasoning effort, creativity, response length, and web search	GPT-5.2 (default), GPT-5.1, GPT-4.1	Balance response quality with cost
Student Instructions	Guidance text students see when they click "Instructions"	Study strategies, academic integrity policies, app tips	Align with course expectations and learning goals
Prompt Template Buttons	One-click conversation starter buttons	"Connect Terms", "Misconception Check," "Study Plan"	Scaffold different types of learning activities
Resources List	Links to download relevant files or URLs	PDF of the course syllabus, Link to course textbook	Create a supportive learning ecology around responsible and effective GenAI use
AI System Prompt	Hidden prompts that define Schema Study's Socratic tutoring behavior	Socratic questioning rules, feedback style, pacing	Adjust pedagogical approach for different courses; Any edits should involve extensive role-play testing

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